

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1864

Elk River 138 kV
August 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1864. The Interconnection Customer proposes a 137.9 MW wind facility interconnecting at the Elk River 138 kV substation in the Pennsylvania area. The request was received on August 20, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 677 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,551,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Redondo-Berthusen 138 kV line
Contingency: Eliza Crk-Ontelaunee Energy Ctr 345 kV
Loading: 148.0% (Rating: 207 MVA)
DFAX: 5.6%
Pre-project loading: 144.3%
2. Facility: Gray-Leake 138 kV
Contingency: Eliza Crk-Ontelaunee Energy Center 345 kV
Loading: 122.8% (Rating: 182 MVA)
DFAX: 5.1%
Pre-project loading: 119.0%
3. Facility: Harvey Couch-Parkland Substation 138 kV
Contingency: Curecanti-Moser Generating Sta 345 kV
Loading: 119.3% (Rating: 329 MVA)
DFAX: 9.2%
Pre-project loading: 115.4%
4. Facility: Parkland-Ft Smith 138 kV
Contingency: Curecanti-Moser Generating Station 345 kV
Loading: 118.2% (Rating: 343 MVA)
DFAX: 9.2%
Pre-project loading: 114.5%
5. Facility: Kingsburg-Ft Smith 138 kV
Contingency: Curecanti-Moser Generating Station 345 kV line
Loading: 116.5% (Rating: 367 MVA)
DFAX: 9.2%
Pre-project loading: 113.0%

6. Facility: Kingsburg-Station Mvalley 138 kV
Contingency: Moser Generating Sta-Curecanti Substation 345 kV
Loading: 115.1% (Rating: 389 MVA)
DFAX: 9.2%
Pre-project loading: 111.9%
7. Facility: Sta Mvalley-Elk Riv 138 kV
Contingency: Moser Generating Sta-Curecanti 345 kV
Loading: 114.4% (Rating: 401 MVA)
DFAX: 9.2%
Pre-project loading: 111.3%
8. Facility: S Reading Substation-Berthusen 138 kV L
Contingency: Eliza Crk-Ontelaunee Energ Ctr 345 kV L
Loading: 110.7% (Rating: 242 MVA)
DFAX: 5.6%
Pre-project loading: 107.5%
9. Facility: Cornelius Delivery No 1-Locbuie 345 kV
Contingency: Owatonna-Moser Generating Station 345 kV
Loading: 107.9% (Rating: 1,163 MVA)
DFAX: 8.0%
Pre-project loading: 107.0%
10. Facility: Lochbuie-Moser Generatin Sta 345 kV
Contingency: Owatonna-Moser Generating Station 345 kV
Loading: 100.3% (Rating: 1,620 MVA)
DFAX: 9.1%
Pre-project loading: 99.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Redondo-Berthusen 138 kV	Rebuild line with higher-capacity conductor	\$3,000,000	26% (\$789,000)
Gray-Leake 138 kV	Rebuild line with higher-capacity conductor	\$5,763,000	100% (\$5,763,000)
Parkland-Harvey Couch 138 kV	Rebuild line with higher-capacity conductor	\$3,101,000	100% (\$3,101,000)
Parkland Substation-Ft Smith 138 kV	Rebuild line with higher-capacity conductor	\$1,712,000	100% (\$1,712,000)
Kingsburg-Ft Smith 138 kV	Rebuild line with higher-capacity conductor	\$2,762,000	100% (\$2,762,000)
Kingsburg-Sta Mvalley 138 kV	Rebuild line with higher-capacity conductor	\$4,153,000	100% (\$4,153,000)
Elk River Substation-Station Mvalley 138 kV	Rebuild line with higher-capacity conductor	\$7,652,000	100% (\$7,652,000)
S Readig Substation-Berthusen 138 kV	Rebuild line with higher-capacity conductor	\$1,079,000	26% (\$284,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cornelius Delivery No 1-Lochuie 345 kV	Reconductor line	\$9,647,000	2% (\$196,000)
Lochuie-Moser Generatin Sta 345 kV	Reconductor line	\$7,860,000	2% (\$139,000)
	Total allocated to project		\$26,551,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.